Sahaj Bhadja

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Summary

WMS Software Engineer with 2+ years of experience building **SQL/Python systems**, **operational integrations**, automated delivery workflows, and **reliable data-intensive software** across production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for operational analytics workflows supporting 500+ users reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable operational software workflows at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **maintainable integrations**, indexing, caching, pagination, and filtered payloads while debugging cross-platform services across Windows, Linux, and macOS production environments reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **resource-management platform**, defining workflows for allocation, customers, billing, permissions, notifications, and cross-location record handling with real-time updates across distributed operational locations reliably.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, Azure monitoring, automated tests, and Docker/Jenkins deployments across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms operational workflows** across 50K+ booking records, integrating pricing, scheduling, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.
- Reduced manual refund handling by approximately 70% through **payment and dispute automation**, implementing webhook processing, validation rules, real-time status synchronization, and traceable workflows across Android, iOS, and web applications.

Skills

Languages: SQL, Python, Java, TypeScript, JavaScript, C++, Bash
Databases: PostgreSQL, MySQL, MongoDB, AWS RDS, Azure SQL, NoSQL, Data Modeling, Schema Design, Query Optimization
Backend & Integration: REST APIs, WebSockets, Microservices, JSON, FastAPI, Node.js, JWT, Caching
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Maven
Operations: Data Validation, Batch Processing, Real-Time Synchronization, Error Handling, Fault Isolation, Monitoring, Technical Documentation, Performance Optimization
Quality: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Production Reliability

Projects

Warehouse Workflow Control - Design Concept
- Designed a **SQL-centered WMS concept** covering inventory locations, license plates, receiving, replenishment, work-in-process, cycle counts, and shipping through validated procedures, auditable transitions, and operational views for warehouse teams.

WMS Integration and Deployment Monitor - Design Concept
- Designed a **Python integration-monitoring concept** for ERP and production-system exchanges, validating messages, tracing failures, replaying recoverable events, documenting changes, and surfacing deployment-check diagnostics for operations support teams.

Chat2SQL | Live
- Extended a **Java and TypeScript database workspace** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for schema exploration, query workflows, structured validation, debugging, and repeatable deployment.